

EVERGLADES NATIONAL PARK HOMESTEAD, FLORIDA

**REPLACE GENERATOR SYSTEMS & COMPONENTS PARKWIDE
PMIS 309068**

STATEMENT OF WORK



NATIONAL PARK SERVICE (NPS)

1.0 PROJECT INTRODUCTION

1.1 Background

Florida experiences frequent events of intermittent power outages due to severe weather. When commercially generated power is not available, Everglades National Park relies on emergency stand-by generators to provide the power to operate its facilities. If power is unavailable for a prolonged period, park facilities must be closed, and residents temporarily relocated due to the unavailability of potable water and sanitation. These generators are subject to relentless south Florida heat and salty ocean air. Most of them are beyond their service life and/or are not functional. Frequent repairs are common, and issues are not known until alternative power is needed and the problem surfaces. Additionally, many of the generator systems are not equipped to fully meet the needs of the location. This project will provide replacement of three (3) standby generators and their components to include, but not limited to, the transfer switches, wiring, conduits, fuel tanks, and fencing, and purchasing of four (4) portable generators.

1.2 Objective

- 1) Provide all labor, materials, and equipment to perform the following work:
 - a) Removal and proper disposal of the existing standby emergency generator systems shown in Attachment A.
 - b) Purchase and installation of three new standby emergency generator systems to replace existing systems shown in Attachment A. New systems will be diesel fueled and sized to provide emergency standby power to the connected facility. Replace all system components including, but not limited to, the transfer switches, wiring, conduits, fuel tanks, fuel lines, and fencing.
- 2) Purchase four new mobile/portable diesel generators equal to the one in Attachment B. The Contractor shall provide the Contracting Officer (CO) with specifications and documentation on their recommended generators for review and approval before purchase is made.

1.3 Location

This project is located at Everglades National Park in Homestead, Florida.

2.0 PROJECT REQUIREMENTS

2.1 Scope of Work and Project Summary/Requirements

The project will consist of removing and replacement of three existing standby generator systems located within Everglades National Park (see Attachment A). The new systems will be diesel fueled and sized to provide adequate emergency power for the connected facility. All necessary ancillary equipment shall be replaced to include, but is not limited to, air filtration systems, generator controls and instrumentation, fuel system, fuel storage tank, cooling system, and engine exhaust system. All transfer switches will be replaced with new automatic switches.

The project scope will include, but not be limited to, the following:

- Perform an assessment of the emergency power needs for the facilities identified in Attachment A based on current industry standards,
- Replacement standby generator systems at the facilities based on those needs,
- Removal and proper disposal of the existing standby emergency generators and components.
- Installation of the new standby emergency generator systems and components, 100KW at Park Headquarters, 75KW at Daniel Beard Center and 30KW at Pine Island Maintenance office.

- Purchase of four new mobile/portable diesel generators.

2.2 Requirements

Replace Three (3) Stand-By Generator Systems

- This line item includes providing all labor, equipment, and materials required to replace three emergency stand-by generator systems identified in **Attachment A**. The scope of work includes purchasing the new generator systems, removing and properly disposing of the existing systems, and installing the new units.
- The replacement effort shall include all components of the existing systems, including but not limited to disconnects, non-service automatic transfer switches, wiring, conduits, fuel tanks, fuel lines, and perimeter fencing. The Park Headquarters building currently uses a manual transfer switch that isolates half of the building; replacement of this switch is required under this contract.
- All new stand-by generator systems must be installed in full compliance with current electrical codes, including but not limited to the following requirements:
 - Service Disconnect (NEC 230.70(A))
 - Overcurrent Protection (NEC 230.91 & 240.21)
 - Transfer Equipment (NEC 702.5)
 - Back feed Protection (NEC 702.6 / NEC 702.5(D) Dec 2020)
 - Grounding and Bonding (NEC 250.24(A)(5))
 - Separately Derived Systems (NEC 250.30)
 - Signage (NEC 702.7)
 - Generator Disconnect (NEC 445.18)
 - Conductors (NEC 215.2 & 215.5)
- Installation Summary by order: Meter →Service disconnect →ATS →Panel.

Purchase Four (4) Mobile/Portable Generators

- Four (4) new mobile/portable diesel generators. Generators shall meet the specifications provided below and mounted on single or dual axel trailer. See Attachment B for additional information on the generators.
- **Prime Range (kW/kVA)** 25-38/25-48
- **Fuel Type** Diesel
- **Frequency** 60 Hz
- **Speed** 1800 RPM
- **Alternator Type** 4-Pole, Rotating-Field
- **Emissions** Tier 4 Final EPA-Certified for Nonroad Applications
- **Sound Levels** 71 dB(A) log average 7 m (23 ft.) with full load at the prime rating
- **Fuel Capacity/Run Time** 88 gal.

STANDARD FEATURES

- Unit-mounted radiator with 45°C (113°F) ambient air capability at the prime rating.
- Superior voltage waveforms from a two-third pitch stator and skewed rotor.
- Approved for use with certified renewable Hydrotreated Vegetable Oil (HVO)/Renewable Diesel (RD) fuels compliant with EN15940/ASTM D975.
- NEMA MG1, IEEE, and ANSI standards compliance for temperature rise and motor starting.

- One-source responsibility for the generating system and accessories.
- A three-year, 3000-hour limited warranty covers all generators set systems and components.
- The trailer shall have minimum of one-year limited warranty.
- Capable of sustained line-to-neutral short-circuit current of up to 300% of the rated current for up to 2 seconds. (IEC 60092-301 short-circuit performance.)
- Sustained short-circuit current enabling downstream circuit breakers to trip without collapsing the alternator field.
- Windings must be vacuum impregnated with polyester resin for dependability and long life.

ENGINE FEATURES

- The generator set engine must be certified by the Environmental Protection Agency (EPA) to conform to Tier 4 Final nonroad emissions regulations.
- Heavy-duty air cleaner with electronic air restrictor indicator switch.
- Cooled Exhaust Gas Regeneration (EGR) and Diesel Oxidation Catalyst (DOC) to meet Tier 4 Final without a Diesel Particulate Filter (DPF).
- Battery disconnect switch.

ENCLOSURE/TANK FEATURES

- Durable steel, sound-attenuating housing with quiet operation of 71 dB(A) log average @ 7 m (23 ft.) with full load at the prime rating.
- Stainless steel hinges and lockable latches on doors.
- 100% secondary containment tank for fuel.
- Single-point lifting eye and four-point tie down system.
- Subbase fuel tank for 25-hour run time with full load at prime rating (minimum).

CUSTOMER CONNECTION PANEL FEATURES

- Decision-Maker® 3500 controller or equivalent with potted circuitry for protection from vibration and debris.
- Externally mounted, recessed emergency stop switch.
- Adjustable trip main line circuit breaker.
- 1 load lug per phase #6-350MCM, 400 amp.
- Remote start/stop capabilities.
- Shore power connection points to front of junction box for block heater and battery charger.

2.3 Project Schedule

The Contractor shall provide the project schedule to the CO before the kick-off meeting. Schedule shall include detailed project phases with deliverable submission dates, review periods, and summary construction schedule with important milestones included for both phases.

The Contractor shall fully develop the construction portion of the project schedule and submit it to the CO before the Preconstruction meeting. Update project schedule monthly throughout contract period until project substantial completion. Status date of each schedule update shall be 10 days before progress payment request date.

Construction work shall not begin until Government approval is obtained for the Final Construction Documents.

The project schedule shall cover entire contract period. The late finish date is the same as the contract's

established completion date. The Contractors shall use Critical Path Method (CPM) with limited use of lead or lag durations between scheduled activities. The Contractor's project schedule shall consist of procurement activities (including mobilization, submittal, and fabrication and delivery of key and long-lead procurement items) and construction activities.

The Contractor's project schedule shall consist of, but not be limited to, the following for each activity:

- Identify activities and numbers numerically (maximum 5-digit). Number activities in increments of 10.
- Concisely describe work represented by activity (maximum 48 characters). Avoid using non-standard abbreviations. Limit work related to each activity to one work trade.
- Activity duration shall be in whole working days with a maximum duration 15 workdays each, unless otherwise approved by CO, except for non-construction activities including mobilization, shop drawing and sample submittals, fabrication of materials, delivery of materials and equipment, and concrete curing.

The Contractor shall ensure the project schedule includes their own work and the Subcontractor's work. The schedule shall show sequence and interdependence of activities required for complete performance of work and shall be cost and resource loaded. The Contractor shall ensure work sequences are logical and the project schedule shows a coordinated work plan. The proposed durations assigned to each activity shall be the Contractor's best estimate of time required to complete activity, considering scope and resources planned. Resource loading of each activity shall list personnel by labor category and equipment type, and capacity proposed to complete activity in duration shown. Include permit requirements and constraints. Consider and include seasonal weather conditions in planning and scheduling of work influenced by high or low ambient temperatures and wind and/or precipitation, to ensure completion of work within contract time.

Project Schedule Updates: Monthly updates of the project schedule shall be an integral part and basic element of the estimate upon which the progress payments shall be made under this contract. If, in the judgment of the CO, the Contractor fails or refuses to provide schedule of values and complete schedule update or revision as specified hereinafter, the Contractor shall be deemed to have not provided the required information upon which progress payment may be made and shall be subject to retainage of a portion of the payment. As part of the monthly update, include the following in a report:

- Brief description of actual progress made during update period.
- Actual and potential delaying activities.
- Impediments to progress.
- Issues related to inclement weather.
- Progress toward established milestones.
- Project float.
- Brief description of work anticipated to be performed in the next month.
- Identify minor revisions to schedule for evaluation and accepted or rejected.

As work progresses, indicate Actual Completion percentage for each activity. If schedule update shows late finish date after contract completion date, at a minimum, include the following in narrative with submission:

- Known delays.

- Actions taken to get back on schedule.
- Pending modifications.
- Impediments or constraints affecting progress.

Progress Payments: Update currently accepted construction schedule monthly. If the Contractor fails to provide monthly updates or revisions to the currently accepted construction schedule, a portion of monthly payment may be retained until updates/revisions are made.

Time Impact Analysis for Contract Modifications, Changes, Delays, and Contractor Requests

Requirements: When contract modifications or changes are initiated, delays are experienced, or the project schedule is revised, the Contractor shall submit to the CO a written time impact analysis illustrating influence of each modification, change, delay, or Contractor request on the contract time.

- **Time Extensions:** Activity delays shall not automatically mean an extension of contract time is warranted or due to the Contractor. It is possible that modification, change, or delay will not affect existing critical activities or cause non-critical activities to become critical. A modification, change, or delay may result in only absorbing a part of the available total float that may exist within an activity chain of the project schedule, thereby not affecting the contract time. Time extensions will be granted according with terms of contract.
- **Float:** Float is not for exclusive use or benefit for NPS or the Contractor. Extension of contract time is granted only to extend the equitable time adjustments to activity or activities affected by modification, change, or delay exceeds total (positive or zero) float available on a particular activity.

2.4 Construction Requirements

Environmental Requirements

More stringent environmental requirements may need to be met for this project. Refer to Part 3 Performance Requirements. General requirements:

- **Site Disturbance:** Minimize construction limits of the project to reduce impact on site.
- **Toxic Chemicals:** Avoid materials that can leach toxic chemicals into ground water. Do not allow toxic chemicals to enter sewers or storm drains or contaminate land or any body of water.
- **Soil Erosion:** Protect against erosion and topsoil depletion.
- **Habitats:** Protect natural habitats and ecological systems on facility sites.

Waste Management: Employ processes ensuring minimal waste output and in landfills. Recycling is a requirement of this project. Legally dispose materials unsuitable for recycling at public or private dumping areas outside the park.

The Contractor shall designate an on-site party (or parties) responsible for instructing workers and overseeing environmental requirements of this project. Distribute copies of environmental requirements to Job Site Foreman and each Subcontractor.

Waste Management Plan: Prior to the scheduled Preconstruction meeting, the Contractor shall submit a draft Waste Management Plan to the CO for approval. The Contractor shall develop and implement it according to ASTM E1609 and as required in contract documents. Plan shall include, but not be limited to:

- List of recycling facilities, reuse facilities, municipal solid waste landfills, and other disposal area(s) to be used. Include name, location, and phone number.
- List of proposed materials to be reused or recycled.
- List of materials that cannot be recycled or reused with explanation or justification.
- Storage and collection methods of waste and recyclables, handling procedures, and means of keeping recyclables free of contamination.
- Describe means of transporting recyclable materials and estimate frequency of emptying bins.
- Identity of person or persons responsible for ensuring project recycling goals are achieved.
- Revise and resubmit Plan per CO. Approval of the Contractor's Plan does not relieve the Contractor of complying with applicable environmental regulations.

Progress Documentation: Supplemental to Waste Management Plan, document solid waste disposal, diversion, and cost/revenue analysis, and submit completed worksheet monthly. See Project Waste Management Plan Worksheet Sample, included with Division 1 Specifications, and report totals to date for column headings.

Prior to commencement of work, schedule and conduct a meeting with the CO. Discuss proposed Waste Management Plan and develop mutual understanding of details of environmental protection.

Minimizing Construction Impacts

Implement following mitigation measures to minimize impacts of construction activities:

Exotic Vegetation and Noxious Weeds: Prevent introduction and minimize spread of exotic vegetation and noxious weeds. Incorporate the following procedures:

- Obtain fill, rock, or additional topsoil needed from a government-approved source.
- Use approved native seeds and/or plants to revegetate the areas disturbed by construction.

Special Status Species: To protect unknown or undiscovered threatened, endangered, or special status species, construction contract will include provisions for these discoveries. The Contractor should contact Park for direction to evaluate special species at the site.

Visitor Experience: Incorporate following mitigation measures into action alternatives to minimize impacts of construction activities on visitor experience:

Air and Water Pollution Control: Air quality impacts are expected to be temporary and localized. To minimize these impacts, take the following actions:

- To reduce tailpipe emissions, do not leave construction equipment idling longer than necessary for safety and mechanical reasons.
- To reduce construction dust, apply water to problem areas. Limit equipment to fenced project area to minimize soil disturbance and dust generation.
- Take necessary reasonable measures to reduce air and water pollution with material or equipment used during construction. Keep volatile wastes in covered containers.

Staging Areas: Limit construction sites to the smallest feasible area. Carefully control ground disturbance and site management to prevent undue damage to vegetation, soils, and archeological resources, and to minimize air, water, soil, and noise pollution.

Staging is limited to an area in or near the construction site for a construction office or trailer. Locate construction equipment and material storage in previously disturbed areas near the construction site.

Obtain CO's approval for staging area location for construction equipment. Once construction is complete, return staging areas to pre-construction conditions. Consult with the CO and develop standards and methods for determining standards.

Structures: Trailers, Storage, Field Offices, Staging Area and Sheds: CO to approve location.

Field Office: Structurally sound construction and weather tight. Location determined by NPS before issuing Start Work Notice.

Storage Sheds: If used, provide weathertight sheds or other covered facilities for materials subject to weather damage and remove temporary water, sewer, sanitary facilities and electric utilities upon project completion. Remove temporary utility connections.

Construction Zones: Before construction activity, use construction barrier, plastic, or portable fencing, approved by CO, to define construction zone, minimum area required for construction. Clearly state protection measures in construction specifications and instruct workers to avoid conducting activities beyond the construction zone.

Protection of Public: Use fence, barricade, or other means to block immediate work area and prevent unauthorized entry. Erect and maintain fencing, barricades, lights, signals, and warning signs according to current version of Manual on Uniform Traffic Control Devices (MUTCD).

The Contractor shall protect tree trunks and root systems of trees in, or adjacent to, work areas.

Parking of Construction Vehicles: Limit to existing roads in legally designated areas and within approved staging area(s).

2.5 Submittals During Construction

Submittals:

Action Submittals: Written, graphic information, and physical samples that require Government's responsive action.

- **Product Data:** Collect information in single submittal for each element of construction and type of product or equipment.
- **Construction Materials:** The Contractor is encouraged to submit, for approval, products made of recycled or environmentally responsible material. NPS will make every effort to approve these materials.

Informational Submittals: Written information not requiring Government's responsive action. Submittals may be rejected for not complying with requirements.

- Prepare and submit Informational Submittals required by individual Specification Sections.
- Informational submittals include, but are not limited to, Certificates, Test reports, Manufacturer's instructions, Division 01 Management Plans, etc.

Contractor's Review: Review each submittal for coordination with other work in the contract and compliance with Construction Documents. Note corrections and field dimensions.

Processing Time: Allow enough time for submittal review and re-submittals. Contract time extensions will not be authorized if submittals are not transmitted with enough time to permit processing, including re-submittals.

1. **Action Submittals:** Allow 30 days for initial review of each submittal. Allow additional time if coordination with subsequent submittals is required.
2. **Informational Submittals:** Allow 10 days for review of each submittal.

Submittals: Identify and incorporate information in each submittal file as follows:

1. **Transmittal Form (CM-16):** Utilize [NPS Transmittal Form](#) for all submittals. No action will be taken on a submittal item unless accompanied by this Transmittal Form.

CO's (CO) Review

Action Submittals: CO will review each submittal, generate comments on corrections or modifications required, and indicate appropriate action on Transmittal Form in one of three ways:

1. **Approved:** Acceptable with no corrections required.
2. **Approved with Notations:** Minor corrections or clarifications required. Comments are clear and no further review required. The Contractor shall address review comments when proceeding with work.
3. **Disapproved - Resubmit:** Rejected for not being in accordance with contract or requiring major corrections or clarifications. CO will identify reasons for disapproval. The Contractor shall revise and resubmit with changes clearly identified.

Informational Submittals: CO will review submittals and accept or reject.

2.6 Quality Control

The Contractor is responsible for quality of work. Independent testing laboratory is responsible for testing. Inspect and test work as needed to ensure quality of materials, workmanship, construction, and finish, and to ensure functional performance follows applicable specifications and drawings.

The Contractor accepts ultimate responsibility, liability and duty to control quality of services provided to the Government as stipulated in FAR Part 36.609-2. The Contractor shall be responsible for professional quality, technical accuracy, drawings, specifications, and other services furnished by the Contractor, their Employees, Agents, Assignees, and Subcontractors under the contract. The Contractor shall, without additional compensation, correct or revise any errors or deficiencies in specifications, and other services.

The Contractor shall demonstrate existence of a Quality Control (QC) system used for work submitted to NPS. The Contractor and Subcontractors shall use this QC system and, at a minimum, include the following:

- Coordination of drawings and specifications within each discipline and between disciplines.
- Verification that documents to be submitted are accurate and correct.
- Checklists for internal review of drawings, specifications, calculations, and cost estimates.
 - Required documents for each submission are included and complete per contract requirements.
 - Required documents have been through a QC review. NPS requires documentation and submittal of QC work performed.

Quality Control Plan: Prior to Preconstruction meeting, submit written CQC plan for approval.

If plan requires revisions or corrections, the Contractor shall resubmit plan within 10 days.

Government reserves right to require changes to the plan during contract period as necessary to obtain quality specified.

No change to the approved plan may be made without written concurrence by CO.

The plan shall include:

- A list of personnel responsible for quality control and assigned duties. Include each person's qualifications.
- A copy of a letter of direction to CQC Supervisor outlining assigned duties.
- Names, qualifications, and descriptions of laboratories to perform sampling and testing, and samples of proposed report forms.
- Methods of performing, documenting, and enforcing quality control of work.
- Methods of monitoring and controlling environmental pollution and contamination as required by regulations and laws.

Weekly Reports: Submit and include inspections and tests on first workday following date covered by report. Use Daily Test Report Information Report Sheet and CMR & CQC Daily Report forms.

Working Hour Restrictions: Limit work and deliveries to weekday hours of 7:00 AM to 3:30 PM unless otherwise approved by CO. No work shall occur on National Park Service (NPS) holidays or weekends without prior approval.

Temporary Services: Temporary materials may be new or used but must be adequate in capacity for required usage, must not create unsafe conditions, and must not violate requirements of applicable codes and standards. Emergency Generators or temporary power Shall be provided if power interruptions exceed the duration of 30 minutes at any location where services are being rendered.

2.7 Fire Protection Equipment

Observe and enforce standards of fire prevention. No open fires are allowed.

Vehicles and Equipment: Provide one fire extinguisher on each vehicle or piece of equipment. Extinguishers shall have a minimum UL (Underwriters Laboratory) rating of 2-A:10-B:C. A capable and qualified person shall be placed in charge of fire protection. Responsibilities shall include locating and maintaining fire protective equipment and establishing and maintaining safe torch cutting and welding procedures.

Hazard Control: Take necessary precautions to prevent fire during construction. Do not store flammable or combustible liquids in existing structures. Provide adequate ventilation during use of volatile or noxious substances.

Spark Arresters: Equip gasoline or diesel-powered equipment, used in potential forest or grass fire locations, with spark arresters approved by U. S. Forest Service. CO will issue written determinations of areas and periods of potential fire hazard. Locate internal combustion equipment so exhausts discharge well away from combustible materials. Locate service areas that are a minimum of 50 feet from buildings. Shut down equipment before refueling.

Smoking and Tobacco Use: Smoking within buildings or temporary storage sheds is prohibited.

Welding: Cutting by torch or welding shall be performed only when adequate fire protection is provided. Contractor shall submit hot work permit for tasks such as welding or using cutting torches prior to an event. COR will provide hot work permits to contractor.

2.8 Temporary Utilities

Electricity and Lighting: The Contractor shall make arrangements with the utility company for metered connection to existing utility and pay costs. Coordinate work with CO. Temporary electrical and work shall meet requirements of current version of NFPA 70 NEC Article 590. When temporary connections are removed, contractor shall restore existing utility services to original condition. Contractor must notify COR 48 hours prior to any disruption of power to facilities or transfers of power.

Heating and Cooling: Furnish temporary heating and cooling. Do not use permanent heating and cooling system without written authorization from CO. When permanent heating and cooling system is approved for use as temporary heating and cooling, pay costs until final acceptance. Install new filters before final acceptance. Equipment warranties start on date of final acceptance.

2.9 Access

Coordinate construction efforts with CO towards minimal impact to the work of NPS personnel and to the visiting public.

During construction of scheduled facilities, the Contractor shall have continuous access to the site.

2.10 Preservation of Adjacent Features

Confine operations to work limits of project. Prevent damage to natural surroundings. Restore damaged areas outside work limits, repair or replace damaged trees and plants, at no additional expense to NPS.

Provide temporary barriers to protect existing trees, plants, and root zones that will remain in place. Do not remove, injure, or destroy trees or other plants without prior approval. Consult with CO and remove roots and branches interfering with construction. To prevent damage, carefully supervise excavation, grading, backfilling, and other construction operations near trees and plants.

Existing Utilities:

Investigate existing utility structure and condition to include sizing, connections, conflicts, and impending upgrades or relocations, if any. The Contractor shall contact utility companies and/or NPS-owned utility personnel to assess capabilities and conditions requirements.

Identify deficiencies and failing infrastructure as well as documentation, timelines, procedures, and coordination as early as possible. Clearly outlined names, emails, phone records, communication forms.

Minimize service disruptions. The Contractor shall coordinate with local utility companies. Arrange disruptions at least 48 hours in advance with NPS and get approval from CO.

Construction procedures shall include preventing accidental disruptions to facilities outside project limits by investigating existing utilities and protection during construction. Accidental disruptions shall be remedied at no cost to NPS.

2.11 Accident Prevention

Accident Prevention Program: Prior to Preconstruction Meeting, submit accident prevention program. Program must be accepted by NPS before work on site begins and shall comply with Occupational Safety and Health Administration (OSHA) and project requirements. Include:

- Name of supervisor responsible for carrying out the program.
- Monthly safety meetings.
- First aid procedures.
- Outline of each phase of work, hazards associated with each phase, and methods proposed to ensure property protection and safety of the public, NPS staff, and the Contractor and Subcontractor's employees.
- Training
- Planning for possible emergency situations.
- Housekeeping and fire protection.

Accident Reporting: Report:

- Accidents defined as death
- Occupational disease
- Traumatic injury to the Contractor's personnel, NPS employees, or the public
- Property damages in excess of \$100
- Fires within seven days.

Complete DSC Contractor Accident/Property Damage Report (CM-22) and forward to CO.

Quality Assurance: Ensure employees are physically qualified to perform assigned duties safely. Do not allow employees to work if their abilities are impaired. Operators of equipment shall be able to understand signs, signals and operating instructions, and capable of operating such equipment.

Accident Prevention Products: Provide:

- First aid facilities.
- Personnel protective equipment shall meet requirements of NIOSH and MSHA (National Institute for Occupational Safety and Health and Mine Safety and Health Administration).
- Emergency instructions, including telephone numbers and reporting instructions for ambulance, physician, hospital, fire department, and park police. Place in conspicuous locations at work site.
- Adequate egresses at all times according to Standard for Safeguarding Construction, Alteration, and Demolition Operations (NFPA 241).
- Hard hats for employees and for up to six visitors.
- Designate and post signs in hard hat areas.

Training: Provide training for first aid and hazardous material handling and storage.

2.12 Temporary Controls

Housekeeping: Keep project neat, orderly, and in safe conditions.

Weather Protection: When inclement weather is expected, provide temporary protection for areas where roofing, siding, windows, doors, or other enclosing elements have been removed or have not been installed. Inspect protective coverings frequently to ensure proper function.

Hurricane season begins June 1 and ends November 30. If a hurricane threatens area, the park will be secured and closed at least 24 hours before anticipated storm landfall. The Contractor will be responsible for securing their job site in accordance with the hurricane safety plan

2.13 Project Close-out

Posted Operating Instructions: Attach operating instructions on or post adjacent to equipment. Include procedures for wiring diagrams, control diagrams, control sequence, start-up adjustment, operation, lubrication, shutdown, and safety precautions in case of equipment failure and other items of instruction recommended by manufacturer.

Cleaning: Before scheduling final inspection, remove tools, equipment, surplus materials, and rubbish. For surfaces damaged due to work of this contract, restore or refinish to original condition. Remove grease, dirt, stains, foreign materials, and labels from finished surfaces. Thoroughly clean building interiors. Remove construction debris from site. At time of final inspection, project shall be thoroughly clean and ready for use.

Operation and Maintenance (O&M) Data: Prior to final inspection, provide one three-ringed binder with O&M data to CO for review. Data shall include manufacturer's standard literature, equipment data

sheets, vendor-furnished as-built drawings, custom written data not included in manufacturer's standard literature, schedules, warranties, parts lists, test results, and Subcontractor list.

After NPS completes review, incorporate comments. Submit three final sets of O&M data to CO.

System Demonstration and Training: Train designated personnel in adjusting, operating (include seasonal and emergency operations, if applicable), maintaining, and safety requirements of equipment and systems. Instructors shall be thoroughly trained in operating theory and practical O&M work for each type of equipment or system. Sequence of training shall follow approved outline of training guide, e.g. O&M Data. Specify training required in individual sections. If there is no duration listed, provide training of sufficient duration to adequately cover subjects.

Training plan as required by commissioning: For each training session, training plan shall include:

- Dates, start and finish times, and locations.
- Outline of information to be presented.
- Names and qualifications of presenters.
- List of texts and other materials required to support training.

2.14 Substantial Completion and Final Inspection

Submit written certification to CO indicating project, or designated portion of project, is substantially complete, and request final inspection. Upon receiving certification, CO will proceed with inspection within 10 days or advise the Contractor of items preventing project from being designated as substantially complete.

When work is determined substantially complete, CO will prepare a list of deficiencies (Punch List) for corrections before final acceptance and issue a Letter of Substantial Completion. If work is not determined substantially complete, CO will notify the Contractor in writing. After completing work, the Contractor shall resubmit certification and request new final inspection.

Following final inspection, if work is determined substantially complete, CO will prepare list of deficiencies for corrections before final acceptance and issue a Letter of Substantial Completion. The Contractor shall complete work described on list of deficiencies within 30 calendar days, as weather permits. If the Contractor fails to complete the work within this time frame, CO may either replace or correct the work with an appropriate reduction in contract price or charge for re-inspection costs according to the contract's Inspection of Construction clause.

Acceptance of the work - After deficiencies are corrected, CO will issue a Letter of Final Acceptance.

3.0 PERFORMANCE REQUIREMENTS

3.1 Facility Performance

Work performed shall comply with applicable laws, regulations, and NPS policies and guidelines.

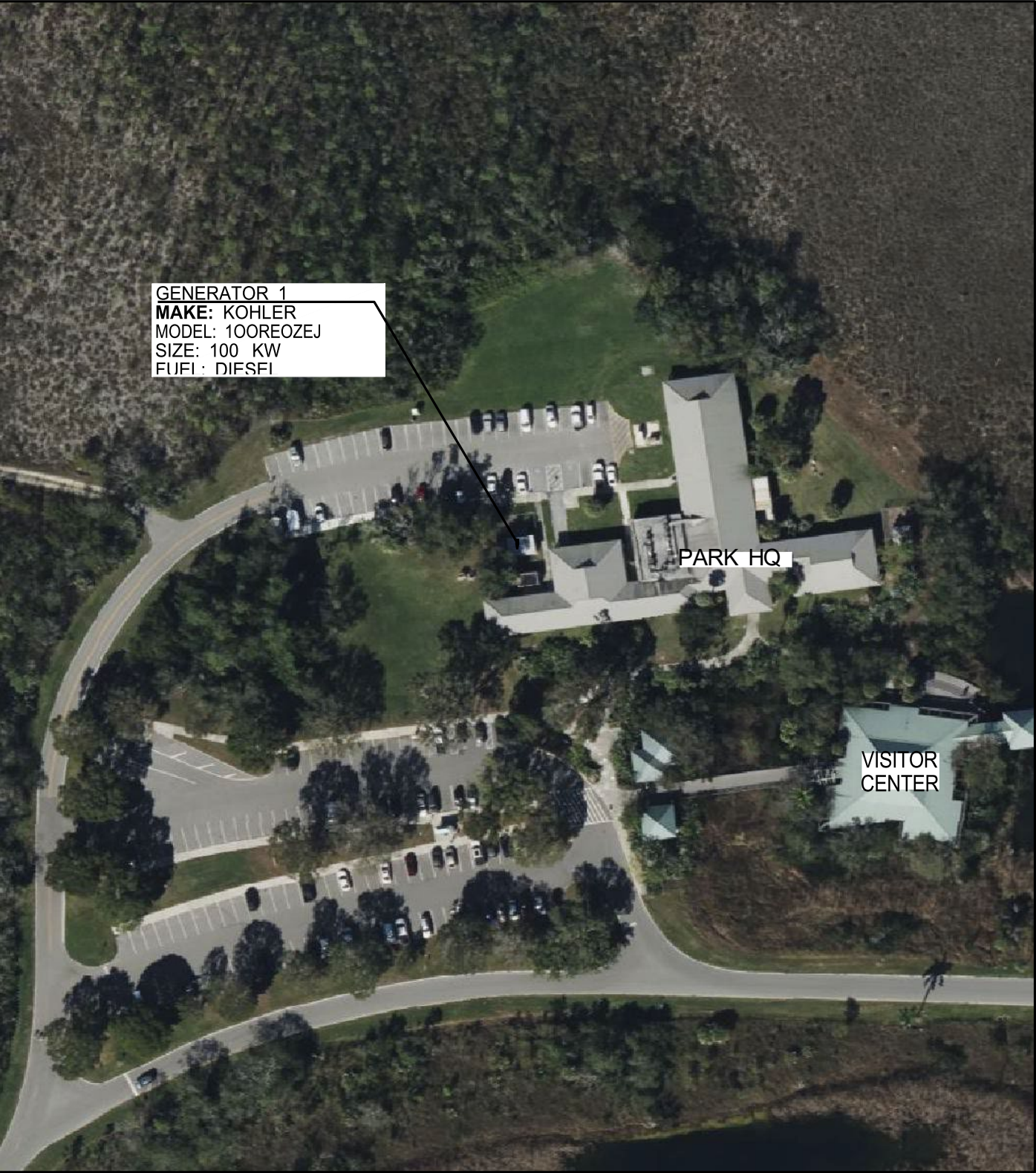
3.2 Elements and Products

In addition to requirements specified in other chapters, provide products and elements that comply with:

- No Substitutions:
 - Use only a specified product (or one of the products).
- Products:
 - Where a product is specified only by a manufacturer name and model number/brand name, the Contractor shall use only that model/brand product.

- Where the properties of a product are specified by description and/or with performance criteria, the Contractor shall use products that comply with description and/or performance criteria.
- Where manufacturers are listed for a particular product, the Contractor shall use a product made by one of those manufacturers that also complies with other requirements.
- Where actual brand name products are not identified, the Contractor shall submit information on recommended products to the CO for approval.
- During Construction, the Contractor shall:
 - Identify actual brand name products used for every product, except commodity products specified by performance or description.
 - Provide test reports showing compliance where a product is specified by performance requirements with test methods and if so specified.
 - Provide manufacturer's product literature for each brand name product.
 - Provide manufacturer's certification that product used on project complies with contract documents.
- Before End of Closeout, the Contractor shall:
 - Provide copies of manufacturer warranties that extend for more than one year after completion.

ATTACHMENT A
STANDBY EMERGENCY GENERATOR PLANS



WORK SUMMARY

- 1. COMPLETE REMOVAL OF STANDBY EMERGENCY GENERATOR SYSTEM AND FUEL TANK, AND FUEL LINES.
- 2. INSTALL NEW DIESEL FUELED STANDBY EMERGENCY GENERATOR SYSTEM, FUEL STORAGE TANK, FUEL LINES.

LOCATION: 25°23'44.77"N 80°35'02.04"W

AutoCAD SHX Text

WORK SUMMARY 1. COMPLETE REMOVAL OF STANDBY EMERGENCY COMPLETE REMOVAL OF STANDBY EMERGENCY GENERATOR SYSTEM AND FUEL TANK, AND FUEL LINES. 2. INSTALL NEW DIESEL FUELED STANDBY INSTALL NEW DIESEL FUELED STANDBY EMERGENCY GENERATOR SYSTEM, FUEL STORAGE TANK, FUEL LINES.





APPROX. 55LF OF TRENCH
FOR TRANSFER SWITCH
CONNECTION

GENERATOR 2
MAKE: KOHLER
MODEL: 30REOZEJ
SIZE: 28 KW
FUEL: DIESEL

WORK SUMMARY

- 1. COMPLETE REMOVAL OF STANDBY EMERGENCY GENERATOR SYSTEM AND FUEL TANK, AND FUEL LINES.
- 2. INSTALL NEW DIESEL FUELED STANDBY EMERGENCY GENERATOR SYSTEM, FUEL STORAGE TANK, FUEL LINES.

LOCATION: 25°23'01.35"N 80°35'39.23"W





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1. COMPLETE REMOVAL OF STANDBY EMERGENCY GENERATOR SYSTEM AND FUEL TANK, AND FUEL LINES.
 2. INSTALL NEW DIESEL FUELED STANDBY EMERGENCY GENERATOR SYSTEM, FUEL STORAGE TANK, FUEL LINES.1

LOCATION: 25°23'17.99"N 80°40'59.04"W

ATTACHEMENT B
PORTABLE/MOBILE GENERATOR INFORMATION



Picture 1: Mobile/Portable Generator Currently in Use



Picture 2: Mobile/Portable Generator Currently in Use



Picture 3: Generator Controls



Picture 4: Connections to Generator